FedCast

A forecaster for the next Federal Reserve rate decision, built to be judged by its process.

Tan Kah Ming · Fullerton · October 2026

About me



Tan Kah Ming

Senior AI, Data and Technology Consultant, seconded to Sovereign Wealth Fund.

Before that

- Data scientist, RSAF Data Office: contributed to RSAF AI Strategy, set up its first digital unit
- Air engineering and weapons logistics officer, where the data-driven habit started (Weibull analysis, predictive maintenance)

Training

M.Eng Aeronautical Engineering, Imperial College London · Graduate Certificate in Predictive Analytics, NUS · CFA Level 3 candidate (February 2027)

Things I have built that models/Applications in

1

Investment analytics, valuation and execution

Investment analytics: AI-assisted performance attribution and idea tracking for an investment team.
Valuation: Machine-learning valuation for illiquid assets.
Execution: Trade workflow automation for investment teams.

2

Data Science

- 4th of 160+ teams (MIT, DoD, USAF) in the 2023 USAF AI Accelerator, predicting pilot flying difficulty from physiology;
- An ammunition recommender (quantile LightGBM) that cut wastage with no shortfall -> Defence Manpower Committee Innovation Award 2022.

3

COVID-19, 2021

Built, in weeks, the system that engaged every home-recovery patient in Singapore, 10,000 people a day, on form.sg, Azure and Zendesk. The lesson that carries over: a system people rely on must say what it knows and what it does not.

What we will cover

1. **The question, and what happened**
   - The 28 October decision, what our outcome says.
2. **Why this approach**
   - Four kinds of signal already exist;
   - Three reasons we built it this way.
3. **How the number is built**
   - Model Architecture
   - App showcase
4. **The LLM analysts**
   - Limitations of model and how we workaround it.
   - The app: the prompts and the results.
5. **How good is the harness**
   - Effectiveness of the harness
6. **What comes next**
   - Follow up
   - QnA
   - Appendix: how the analysts' opinion moves the number; the probit.

The question, the answer, and what happened

28 Oct

the decision being forecast, 2026

67.6%

FedCast, no change: ledger entry #18, nine hours before

Hold

what the Fed did: 3.75 to 4.00%, unchanged

10 to 2

two dissents, both for the +25 bps the specialists leaned toward

How the number was made

Evidence

23 items frozen on 28 October: futures, eight FRED series as first published, Fed statements, minutes, transcript, projections, speeches, 32 years of decisions, the NY Fed natural rate, prediction-market odds.

Classic models, weighted by code

Futures-implied 0.6, the Fed's Taylor rules 0.2, prediction-market odds 0.1, historical base rates 0.1; a 1.5% floor on every outcome. Pooled: **no change 70.9%, hike 26.2%.**

LLM analysts on top

Five specialists read the Fed's words; every claim quoted and verified by code.
Their mean opinion moves 4 points toward a hike:
no change 67.6%, +25 bps 22.4%, +50 bps 7.7%.

What the outcome says

1. We hit the target. So did the market (90/10) and the odds (95/5); at five weeks we do not claim to beat them.

2. The LLM analysts got the direction right, and the vote agreed:

Analyst	Opinion	Reads
4 Communications	+6.0 hike	statements, minutes, transcript, speeches
5 Minutes	+6.2 hike	the minutes only
6 Inflation	+5.8 hike	inflation pre-read, projections
8 Financial conditions	+3.4 hike	2-year gap, breakevens, NFCI
7 Labour market	−1.4 hold	labour pre-read, projections
Mean, one vote each	**+4.0 hike**	two dissents for a hike in a 10 to 2 vote

3. Every reason is a quote code found in the frozen evidence. One decision proves little: it is scored formally (log score, Brier, against the market and two naive baselines) when the post-meeting rate row lands.

Live demo 1: the forecast

1 min

Open

- The app at localhost:8765, Forecast view
- Top bar: when the evidence was pulled, what is new at the sources

Point at

- The two tracks side by side: machine only, and with views
- The meta line: ledger entry, snapshot hash, the specialists' tilt
- The five bars

Say

- This number was made at <time> from evidence frozen at <time>. Everything I show next is how it got there.

LIVE DEMO

Four kinds of signal, and where FedCast sits

Market signals

Fed funds futures (CME FedWatch); prediction markets (Polymarket)

What it is: Prices and crowd odds on each outcome

Strength Futures about 88% right at 30 days, 1994 to 2024; odds move fast

Policy rules

The Taylor family

What it is The rate a formula says is appropriate given inflation and unemployment

Strength Transparent; the Fed's own published benchmarks

Economic signals

Economist surveys; the macro data itself

What it is The median of polled forecasters; the data the committee reads

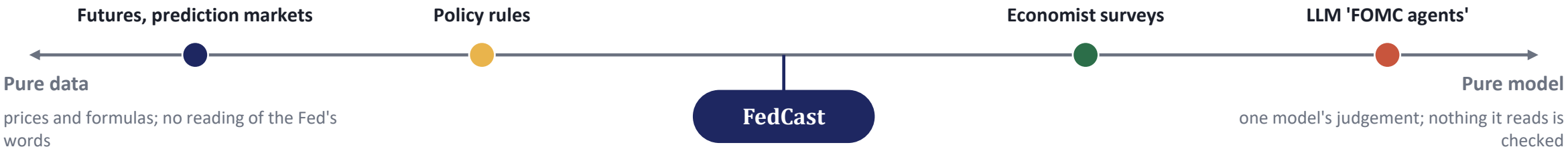
Strength Broad human judgement; the data is the Fed's own input

ML / AI signals

LLM 'FOMC agent' papers (2025)

What it is Models that read the Fed's text and vote

Strength Reported 90%+ accuracy on 2023 to 2024



Takes one input from each area, prices, 32 years of decisions, the Fed's own rules and a language model that must quote what it reads, and combines them by code at fixed weights. Neither a formula nor an oracle.

Sources: Bonini, Huang and Simaan, 'Watching the FedWatch' (232 meetings, May 1994 to March 2024); FedSight AI preprint, December 2025; Takano, Hirano and Nakagawa, arXiv 2511.02469.

Why this approach ?

1 We judge the process, not the luck

- We are at early stages of model building hence **process takes precedence over results**.
- A good process is a repeatable one:
  - Same evidence, same answer?
  - Is every claim quoted?
  - any number from memory?
  - Does a shock move the number the right way?
- Outcomes are logged after each decision and never used to tune

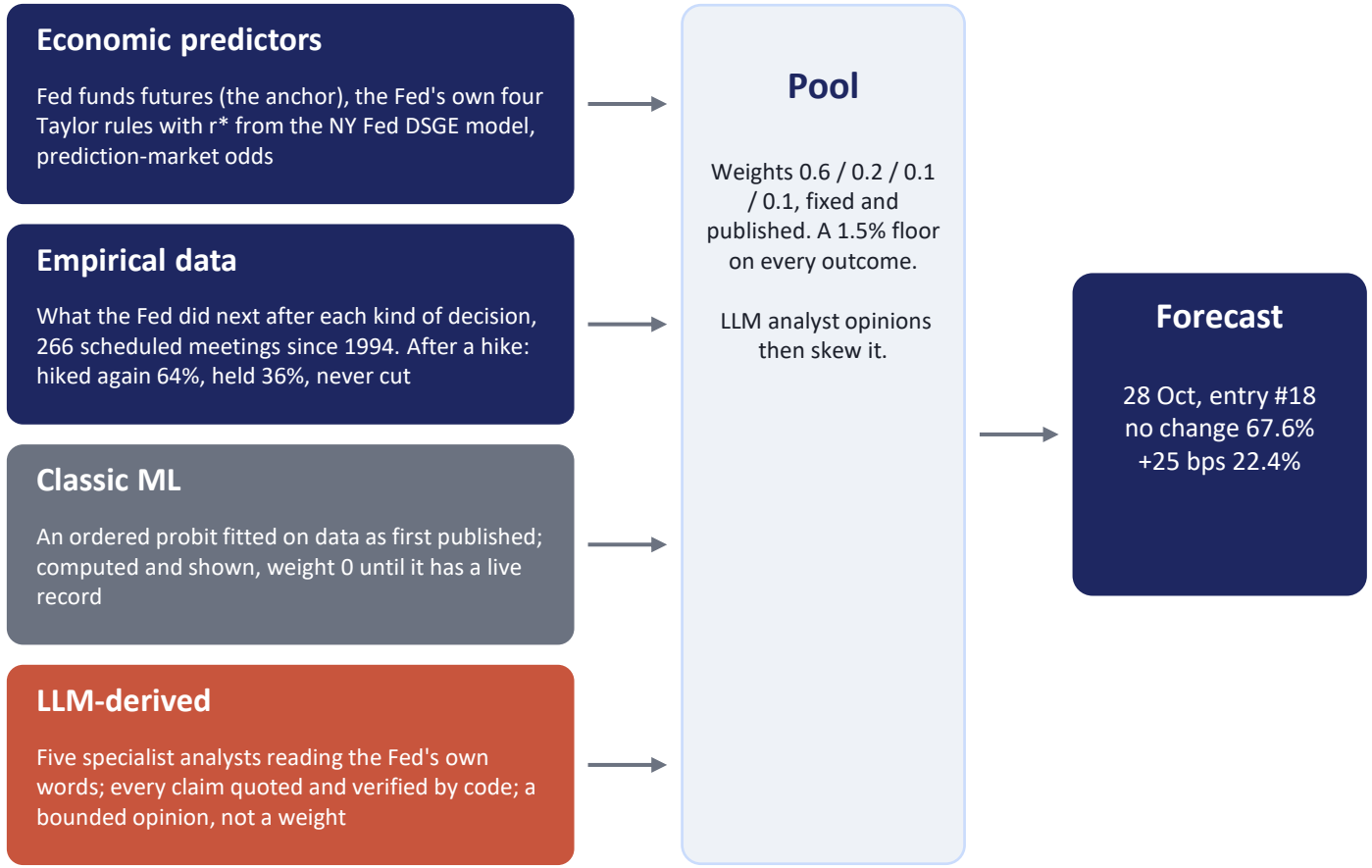
2 We build every step to be auditable

- An analyst that says 'inflation remains elevated' must quote the statement line it read, and code checks the quote is there, character for character
- Every number it uses must exist in the frozen evidence, or **it is flagged as from memory**
- Every model lists the items it read; every ledger entry recomputes from its snapshot. If it is not quoted, it does not count
- We believe that this method allows us to **use less expensive models** as well

3 A harness that compounds

- Since everything is quotable and the LLM analysts can read only the evidence, so a wrong call is a traceable wrong call: we see which quote led where.
- The harness is repeatable, so it goes wrong the same way every time; a fix can be tested, and the drift table shows it stayed fixed
- That is the virtuous cycle: audit, fix, re-measure. The ladder, the scorecard and the drift table later in the deck are that cycle

Model architecture: Ensemble model



The chain, in order

Evidence — 23 items, frozen and fingerprinted, dated in US time, data as first published.

Signals — Numbers computed by code: inflation gap, labour slack, financial conditions, statement tone, base rates, projections.

Pool — Four families at fixed weights; a 1.5% floor; then the specialists' opinion.

Analysts — Analysts 4 to 8 enter the number. Analysts 1 to 3 are controls.

You — Views move the number by at most 10 points and expire; watch-outs are answered by id; documents and odds become evidence.

Ledger — Append-only, hash-chained; every entry recomputes from its snapshot.

The arithmetic: from five models to one number

Model	What it reads	−25	no change	+25	+50	weight
Market-implied	The November fed-funds futures price	0	90	10	0	0.6
Taylor family	Core PCE inflation, unemployment, the NY Fed natural rate, through the Fed's own four policy rules	4.5	26.9	44.6	23.8	0.2
Prediction market	Polymarket odds on each outcome, 26 October	0.2	94.9	4.5	0.4	0.1
Historical base rates	What the Fed did at the 49 meetings since 1994 that followed a hike	0	34.7	52.2	13.1	0.1
Ordered probit v2	Nine macro features, as first published; fitted on decisions since 2000	0	24.2	63.8	12.0	0
Pooled	Weighted average, then the 1.5% floor	**1.5**	**70.9**	**20.2**	**6.0**	
After pooling	**Specialists' opinion: a modifier on the pooled number, not a model in the pool.** Mean of analysts 4 to 8, +4.0, hawkish: 4.0 points of probability moved toward a hike, capped at 10.	−0.3	−3.3	+2.2	+1.7	—
Machine forecast	Ledger entry #18, 28 October	**1.2**	**67.6**	**22.4**	**7.7**	

Weights are based on judgement

Logged as amendments, never fitted. Ideally we would backtest the appropriate weights; with eight decisions a year there is no record to fit on yet, so live scores from 28 October onward decide them, by amendment.

Live demo 2: the evidence graph and the pool

3 min

Open

- Evidence graph
- Click the Market-implied box
- Click the Pool box
- Methodology, scroll to the worked example

Point at

- Market-implied: the futures price, 100 minus price, the expected change, the 90/10 split
- Pool: the four weights, the pooled line, the specialists' tilt line, Machine track
- Any evidence item: the frozen text and the link to the original

Say

- Blue is in the number, grey is computed but not in it. Click a model and you see only what it read.
- The language model's opinion is the one line between Pooled and Machine track. It can tilt, it cannot write.

LIVE DEMO

Working around the model's limits

Two things a language model cannot be made to stop doing. Neither is removed; each is designed around, and the design is measured on every snapshot.

LIMIT

It answers from memory, and differently each time

It knows the Fed from its training and will use that unless stopped; ask the same question twice and the number moves.

WHAT WE DO

- **Break the question down:** instead of one big 'what will the Fed do', each analyst gets small fixed questions with a fixed answer shape, like a multiple-choice and short-answer paper. Less room to wander, so less variation between runs.
- **Evidence only:** the instruction says use only the evidence; every claim must carry a verbatim quote from a named item, so a remembered fact has nowhere to stand.
- **Measured:** every number in the answer is matched to the snapshot by code; five identical runs per analyst, and the spread between them must stay within 2 points.

LIMIT

It can invent its own support

A fluent quote or figure that was never in the document.

WHAT WE DO

- **Check by code, not by a model:** every quote is matched to its item character for character; a quote stitched from two places fails.
- **Bounds:** every watch-out answered by id; the opinion clipped to −10 to +10.
- **One repair round:** the verifier's report goes back once; both answers are kept; what still fails stays unverified, never hidden.

LLM breakdown: the group of analysts

CONTROLS — BUILD THE LADDER

1 — Naive

The question and the date. Nothing else.

Intent: what the model does on its own, from memory. Its output also forms the baseline for every other step.

2 — Evidence shown, nothing verified

Every document and pre-read, 128,000 characters. Free-form answer.

Intent: to observe what evidence alone fixes (memory), and what it does not (variance; nothing checkable).

3 — Generalist with verifier

Same evidence; a fixed answer shape, a quote per claim, code checks every quote and number.

Intent: isolate the value of the structure and verification.

SPECIALISTS · THE FIVE THAT ENTER THE NUMBER — IN THE FORECAST

Analyst 3's shape and verifier, with one change: each reader gets **one slice of evidence**. A judgement is attributable to its slice, and a dissent survives instead of being averaged away.

4 — Communications

Both statements, the minutes, the transcript, the speeches, the tone pre-read.
Intent: what the committee is signalling.

5 — Minutes digest

The minutes only.
Intent: the committee's own account of its debate, read in isolation.

6 — Inflation

Projections table, both statements, the inflation pre-read.
Intent: is inflation pulling toward a hike.

7 — Labour market

Projections table, both statements, the labour pre-read.
Intent: is labour pulling the other way.

8 — Financial conditions

Both statements, the financial-conditions and base-rate pre-reads.
Intent: is the market or stress already doing the committee's work.

Each specialist is asked one question about its own slice (above), not how likely each outcome is. It returns claims, each with a verbatim quote and item id, and an opinion from -10 (strongly dovish) to +10 (strongly hawkish). Code averages the five opinions and moves that many points of probability from hold toward a hike (positive) or a cut (negative), capped at 10.

Every analyst runs five times on the same snapshot. For analysts 1 to 3 those runs are not forecasts: they build the benchmark that 4 to 8 are measured against.

1 sets the baseline: variance of the probabilities across runs; numbers from memory (all of them); nothing quoted.

2 tests immutable evidence: numbers from memory fall to zero; the variance does not; nothing can be checked.

3 tests the answer shape and code verification: variance falls under 1 point; every claim quoted and checked.

Live demo 3: what each analyst is asked and given

2 min

Open

- Analyst prompts view
- Analyst 1: open the full prompt
- Analyst 6: open the full prompt

Point at

- Analyst 1: 195 characters, no evidence
- Analyst 6: the system prompt, the enforced answer schema, the inflation pre-read, the item ids
- The prompt hash: five repeats share it

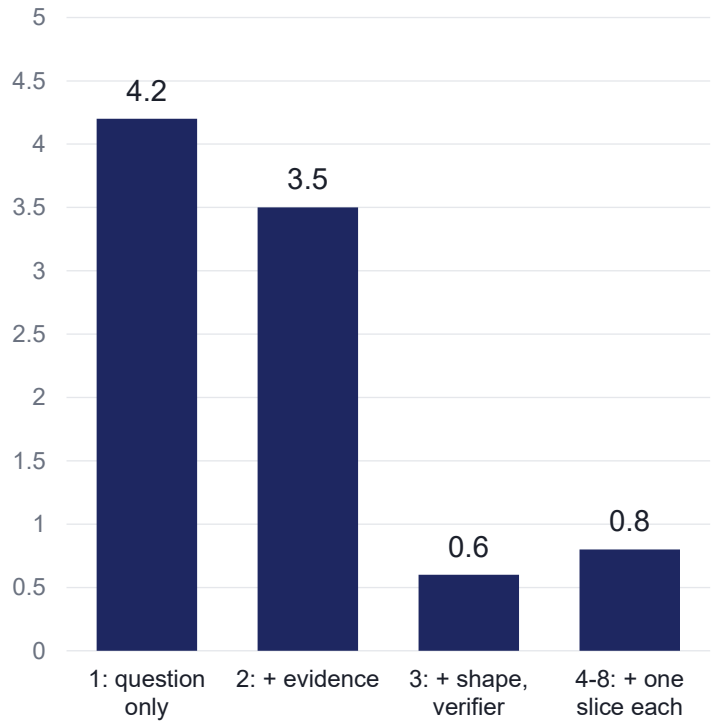
Say

- If it is not on this page, the analyst does not get it.
- Same model. The only difference between analyst 1 and analyst 6 is this page.

LIVE DEMO

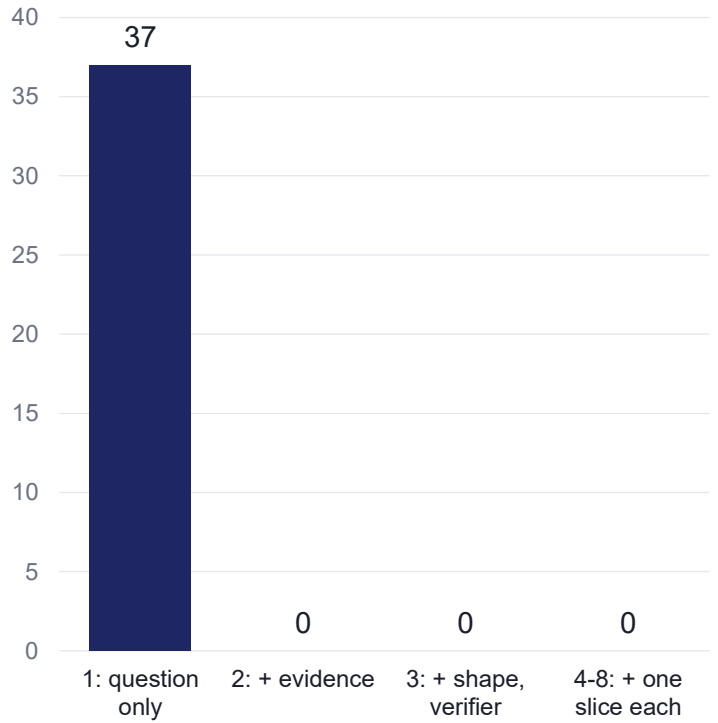
How good is the harness: what each step bought

Repeatability: variance between identical runs (pp)



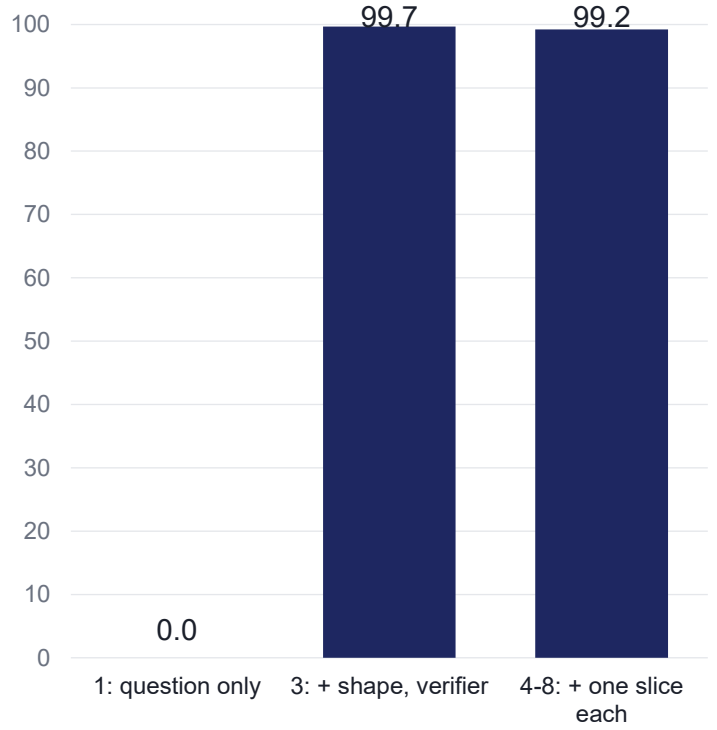
Largest standard deviation across five runs, mean of four snapshots. Pass mark 2 pp. Analysts 3 to 8: standard deviation of the score, in points of probability.

Cited numbers from memory (count)



Numbers in the answer that exist nowhere in the frozen evidence, summed over five runs. Pass mark: exactly 0.

Claims with a verbatim quote (%)



Share of claims whose quote code found character for character in the item named. Pass mark 95%. Analyst 2 is not shown: it quotes nothing, so it cannot be measured.

The step from analyst 2 to analyst 3 is the harness: a fixed answer shape with a quote per claim, checked by code. Specialists keep both and add attribution. What they concluded, four snapshots: analyst 1 a cut at 57 to 67%; analyst 2 +25 bps at 50 to 57%; analyst 3 +5.8 to +7.0 hawkish; specialists +4.0 to +4.4, labour the lone non-hawk.

LLM analysts: outcome

Each specialist runs five times on the same frozen snapshot. The five answers must sit within the variance pass mark (2 points), every claim must carry a quote that code finds in the evidence, character for character, and no number may come from memory. Only then does the opinion count toward the number.

25 runs
5 specialists × 5 identical runs

25 prompts
one slice each, hashed

263 claims
every one with a verbatim quote

263 verified
100% from evidence
0 numbers from memory

5 opinions

WHAT THE FIVE SAID, AND WHAT IT DID TO THE NUMBER

Communications	**+6.0 hike**
Minutes	**+6.2 hike**
Inflation	**+5.8 hike**
Financial conditions	**+3.4 hike**
Labour market	**−1.4 hold**
Equal-weight mean	**+4.0**

Four say hike, one says hold: +4.0. **+4.0 moves 4 points of probability to the hike side.** Each bucket is scaled by a factor rising from the cut end to the hike end, so the hikes gain 4.0 and hold plus the cuts lose 4.0:

Outcome	Before	After	Change
−50 bps	1.5	1.1	−0.45
−25 bps	1.5	1.2	−0.27
no change	70.9	67.6	−3.28
+25 bps	20.2	22.4	+2.24
+50 bps	6.0	7.7	+1.76

ONE VERIFIED CLAIM, AS RECORDED · ANALYST 4, COMMUNICATIONS, RUN 1 OF 5

Claim The September statement stressed elevated inflation and a resolve to return to target on a timelier basis.

"Inflation remains elevated. Today's policy action will support a timelier return to the Committee's 2 percent goal. The Committee will deliver price stability."

Item **fed.statement.2026-09-16** · quote found by code, character for character · opinion hawkish · 12 of 12 claims in this run verified

Live demo 4: what the analysts concluded

2 min

Open

- Analyst results view
- Analyst 1 (naive) card
- Analyst 4 (communications) card

Point at

- Analyst 1: cut 25 at 57%, numbers from memory flagged
- Analyst 4: the opinion, grounded, leaked, watch-outs badges; one claim with its quote and the 'quote verified' mark
- Hover a badge for its definition

Say

- Every claim you see has a quote that code found character for character in the item named.
- The naive analyst still thinks the Fed is cutting. Same model, no evidence.

LIVE DEMO

Live demo 5: the scorecard and the drift table

2 min

Open

- Scorecard view
- The snapshot dropdown: pick 24 October, then 26 October
- Expand 'what was checked' under Shock direction
- Scroll to the over-time table

Point at

- The grid: red in the control columns, green in the judged columns
- The seven shocks with their deltas
- The over-time table: one row per snapshot, the marks never move

Say

- Analysts 1 and 2 are meant to fail. A red cell on the right would be a defect.
- This is the drift check. Two red cells have already pointed at defects and both are fixed.

LIVE DEMO

Live demo 6: the ledger and replay

1 min

Open

- Ledger view
- Press Verify ledger

Point at

- Every entry with its snapshot hash, spec hash and code version
- The replay result: chain intact, entries under the current spec recompute exactly
- Eight numbered amendments, each with its reason, each a new spec hash pinned in the tests

Say

- Nothing in this ledger can be edited or deleted without the chain breaking.
- Any entry can be recomputed from its snapshot. That is what auditable means here.

LIVE DEMO

What comes next

December is on file: entry #23 for the 9 December meeting, +25 bps 73.6%, no change 16.7%, specialists +4.0. The 28 October call is scored the moment the post-meeting rate row lands.

1 Pre-registered weight sets, scored live

Write down three or four candidate weight sets now (for example market 0.6 / 0.5 / 0.7 with the rest redistributed). Score every set on every live decision with the same log score and Brier. The live set changes only by amendment, on that record. Weights stop being judgement without ever being fitted to the past.

2 Reasoning quality, graded by people

A grading form in the app: a user marks an analyst's answer from 1 to 5 on four questions (did it start from the base rate, argue both sides, say what it was unsure of, say what would prove it wrong). Twenty hand grades become the reference; a model grader is trusted only once it agrees with them. The grade becomes the tenth measured dimension and a gate on the opinion.

3 An events agent for regime shifts

1987, 1998, 2008, 2020: the Fed cut on financial stress, differing from what the models read.

A watch-out agent that scans for bank failures, funding stress and market dislocations and raises a typed watch-out every analyst must answer.

4 Further validation from second model family. (cheaper?)

The same harness, the same snapshots, a different language model.
If the ladder reads the same, the harness is doing the work, not the model.

5 Scale: Other central banks and geographies

ECB, Bank of England, MAS. The chain needs four things: a meeting calendar, a policy-rate series, the bank's own statements and minutes, and a futures or swap curve. The scorecard, the ledger and the analysts transfer unchanged; only the evidence sources change.

6 Improvement: Better classical models

Our probit scored poorly. The cause is the data: 68% of decisions are holds and the cuts bunch into a few episodes, so the model learns hold and misses the rest. Replace it with a model built for that imbalance (class weights, or an ordinal model that shrinks toward the base rate), tested only on decisions it has not seen. About 130 usable decisions, so small models.

Three things to take away

- The harness is the product. Same model, same evidence, same day: naive it wobbles and leaks; harnessed it is stable, quoted and checked.
- Judge the process. Eight decisions a year cannot prove skill; a pre-registered scorecard measured on every snapshot can show discipline and catch drift.
- The ask: scrutinise the scorecard, challenge a quote, and tell us which other decisions deserve the same chain. The 28 October call is on file; the 9 December forecast is too.

Backup: the ladder, four snapshots, five runs each

Dimension (pass mark)	Analyst 1: the question only	Analyst 2: + the evidence	Analyst 3: + answer shape + verifier	Analysts 4-8: + one slice each
Repeatability (≤ 2 pp) largest standard deviation of any outcome probability across 5 identical runs	**3.4 / 4.5 / 4.9 / 4.1 pp, fail**	**3.3 / 2.3 / 4.8 / 3.4 pp, fail**	**0.6 / 0.5 / 0.5 / 0.8 points, pass**	**0.5 / 0.8 / 1.2 / 0.8 points, pass**
Quotes verified (≥ 95%) share of claims whose quote code found in the evidence	**0%, fail**	not measurable: no quotes	**100 / 100 / 100 / 99%, pass**	**100 / 97 / 100 / 100%, pass**
Numbers from memory (exactly 0) numbers in the answer that are not in the snapshot	**36 / 36 / 39 / 38, fail**	**0, pass**	**0, pass**	**0, pass**
What it concluded	Cut 25 bps at 57 to 67%: its memory of the Fed is stale	+25 bps at 50 to 57%: reads the evidence, overshoots the market	+7.0 / +6.6 / +6.4 / +5.8 hawkish	+4.4 / +4.1 / +4.4 / +4.0 hawkish; labour the lone non-hawk

Snapshots of 15, 24, 26 and 28 October 2026. Analysts 3 to 8 report repeatability in points of their opinion because they output an opinion, not a distribution.

Appendix: how the analysts' opinion moves the number

1. The opinion

- Each specialist returns one number from −10 to +10: how many points of probability its evidence says should move toward a hike (positive) or a cut (negative). It is not a probability and not a rate forecast.
- Ten is the cap any single input has on the number, the same as a human view.

2. The average

- Five opinions, one vote each, plain mean. 28 October: +6.0, +6.2, +5.8, +3.4, −1.4; mean +4.0.
- That mean is the number of points that must cross from the hold-or-cut side to the hike side. It is applied after the pool and the floor, by the same mechanism as a human view, and the pool before it is always recorded beside the result.

3. The spread, in words

Number the five outcomes −2, −1, 0, +1, +2 from most dovish to most hawkish. Multiply each bucket by e raised to (steepness × its number), then rescale so the five sum to one. Code searches for the one steepness at which exactly the mean opinion's points cross the line; on 28 October it was 0.153. Each bucket moves in proportion to its own size and its distance from the middle, so no bucket is zeroed and the shape is kept.

28 October, entry #18: mean opinion +4.0

Outcome	Pooled, before	Multiplier	After	Change
−50 bps	1.50	× 0.70	1.05	−0.45
−25 bps	1.50	× 0.82	1.23	−0.27
no change	70.86	× 0.95	67.58	−3.28
+25 bps	20.17	× 1.11	22.41	+2.24
+50 bps	5.97	× 1.29	7.73	+1.76
Hold-or-cut side	**73.86**		**69.86**	**−4.00**
Hike side	**26.14**		**30.14**	**+4.00**

Multiplier = e^(0.153 × position) ÷ 1.0486, where position is −2, −1, 0, +1, +2 and 1.0486 is what the five re-weighted buckets add up to before rescaling. Exactly 4.00 points crossed; the after column is the machine forecast recorded as entry #18.

Appendix: the ordered probit, and why it sits at weight 0

What it is

- An ordered probit: the decision (cut, hold, hike) as a function of signals, with two cutpoints. Fitted on every scheduled decision since August 2000, each read with the data as first published (ALFRED vintages), never today's revised figures.
- Version 1, seven features (inflation gap and momentum, unemployment and its 12-month change, payrolls, the 2-year yield gap, the previous decision). Spec frozen before fitting; single split at 2018. Result: log score −1.14 against −0.71 for a base rate. Failed, reported as such.
- Version 2 adds two features (at the zero bound; the streak of same-direction moves) and shrinks 50% toward the base rate. Written after seeing v1 fail, so its score on 2018 onward is optimistic: a declared contamination caveat.
- Weight 0 until live decisions judge it. Computed and shown on every snapshot, never hidden.

The gate: expanding window, 129 decisions since 2010

Candidate	Log score, all	2010 to 2017	2018 on	Accuracy
Base rate, unconditional	−0.744	−0.550	−0.918	77%
Base rate given the previous decision	−0.584	−0.449	−0.704	81%
Probit v1	−0.788	−0.293	−1.232	78%
Probit v2 (caveated)	−0.469	−0.353	−0.572	80%

For each decision from 2010 on: fit on all decisions before it, predict it. Log score: ln of the probability given to what happened (0 is perfect). The base rate given the previous decision beat the fitted model and joined the pool at 0.1 (amendment #6).

Why not a bigger model

About 130 usable decisions; 68% are holds and the cuts bunch into a few episodes. A big model memorises the episodes and looks good on any test near them. The replacement must handle that imbalance (class weights, or an ordinal model shrunk toward the base rate) and be judged only on decisions it has not seen, the expanding window.

Appendix: historical base rates

What it is

- The simplest model in the pool, and the only one with no parameters: nothing is fitted, nothing can be tuned.
- Take the 266 scheduled decisions since 1994. Group them by what the Fed had done at the previous meeting: a hike, a hold or a cut. Count what it did next. That count is the forecast.
- Today the last decision was a hike, so the model reads the 49 meetings that followed a hike: hold 35%, hike 65%. The hike share is split 80/20 between +25 and +50, the historical share of large moves.
- Why it is in the pool: on the expanding-window test it beat the fitted probit (log score −0.584 against −0.788), so it earned 0.1 (amendment #6). Why not more: it knows nothing about today; it is the prior the other models update.
- Intermeeting moves (28 since 1994) are counted separately and never as a scheduled decision.

What the Fed did next, 266 scheduled meetings since 1994

Previous decision	Cases	Next: cut	Next: hold	Next: hike
A hike (today's case)	**50**	**0%**	**36%**	**64%**
A hold	183	6%	85%	9%
A cut	32	66%	34%	0%
All decisions	266	12%	69%	19%

Read the first row: of the 50 meetings that came right after a hike, the Fed hiked again at 32 and held at 18; it never cut straight after a hike. The pool uses the 49 cases known at the time of the 28 October snapshot.

Why a prior matters

The Fed moves in runs. A hike is usually followed by another hike, a cut by another cut, and a hold by a hold. Any forecast that ignores this starts from the wrong place; the base rate is the place to start, and the market, the rules and the analysts move it from there.